Experimentation Results

What I did

- Pipeline: C_1 finds objects (YOLOv11n), C_2 predicts their paths (AutoBot-Ego style), C_3 plans the ego path (PlanT style). All three are ML models.
- Data: 350 nuScenes scenes, front camera. C_1 is trained in Boston, C_2 and C_3 on other Boston scenes, C_1 is updated on Singapore, and all tests use the same 50 Singapore scenes (1,720 test samples).
- Two old C_1 (seed 0 and seed 1). Each is updated 15 times (100/50/25 % of the data, 5/10/20 epochs, 3 seeds): **30 updates**.
- C_2 and C_3 are trained once in two ways and never retrained during the updates: **old-pipeline training** (learn from the old C_1 output, including its mistakes) and **isolated training** (learn from the ground truth).
- For each update: δ_1 = gain of C_1 (mAP50), Δ_2 = increase of the prediction error, Δ_3 = increase of the planning error at 3 s (metres).

Result 1: every update made C_1 better

All 30 updates increased mAP50 (δ_1 from +0.048 to +0.118; old C_1 0.109 and 0.117, updated C_1 0.166 to 0.235).

Result 2: C_2 and C_3 work (checked against simple baselines)

Model	Error on the test scenes	Simple baseline
C_2 (input: ground truth)	0.77 m	4.48 m (objects stay still)
C_2 (input: C_1 output)	2.39 to 2.95 m	5.39 to 5.85 m
C_3 without the ego path	9.3 to 12.5 m	9.25 m (average path) → dropped
C_3 with the last 1.5 s ego path	3.70 to 4.16 m, collisions 5–8 %	9.25 m, collisions 18 %

C_3 still uses the objects: removing them moves its plan by 0.73 to 1.65 m.

Result 3 (main): a better C_1 harms the C_2 that learned its old mistakes

Number of updates (out of 30) in which the error increased. Three levels, from weak to strong evidence. "Larger than random variation" means larger than the change caused by only replacing the old C_1 with the other old C_1 (0.112 m for prediction, 0.018 m for planning). "Statistically significant" means the increase stays above zero in 95 % of 2,000 random re-samplings of the 50 test scenes.

	Prediction error (C_2)		Planning error (C_3)	
	old-pipeline training	isolated training	old-pipeline training	isolated training
Any increase	**23**	5	18	29
Larger than random variation	**11**	0	13	29
Statistically significant	**3**	0	1	10

- In **30 of 30** updates, the C_2 from old-pipeline training got worse than the C_2 from isolated training. Same input, different training: the harm comes from learning the old mistakes.
- All 11 updates larger than random variation start from the old C_1 with seed 0 (mean $\Delta_2 = +0.165$ m). This C_2 also gets worse (+0.112 m) when the old C_1 with seed 0 is simply replaced by the old C_1 with seed 1, although both have almost the same accuracy (0.009 mAP50 apart). This C_2 reacts to the different mistakes, not to the accuracy.
- Planning changes very little (mean Δ_3 −0.006 to +0.058 m, at most 0.13 m of about 4 m), and the increase is more frequent after isolated training. Learned mistakes cannot cause an increase after isolated training.
- By the simple rule ($\delta_1 > 0$ and $\Delta_2 > 0$ or $\Delta_3 > 0$), entangled enhancement appears in **29 of 30** updates.

Result 4: retraining C_2 and C_3 on the new C_1 lowers both errors

Tested on 2 updates (new-pipeline training). Prediction error 2.75 / 2.61 m → 2.39 / 2.45 m, planning error 4.14 / 4.00 m → 3.71 / 3.70 m. Both are even below the errors before the update.

Result 5: no warning sign found yet (H2 not supported)

Nine measures of how much the output of C_1 and C_2 changes (no ground truth needed) were tested. Correlation here means how well a larger change goes together with a larger harm (1 = perfect, 0 = no relation). The best measure, how far C_2's predictions shift, has a correlation of +0.68 over all 30 updates. For the 15 updates of each old C_1 separately, the correlation is only +0.29 and −0.28. The measure therefore only tells the two old C_1 apart and does not warn about single harmful updates. The accuracy gain δ_1 does not predict the harm either (correlation +0.04). Hint for the next step: the old C_1 whose C_2 got worse when the old C_1 was replaced is also the one whose updates caused harm.

All 30 updates

Changes after each update, in metres for Δ_2 and Δ_3. Shaded cells are increases larger than random variation. The last column names the errors with a statistically significant increase after old-pipeline training.

#	Old C_1 seed	Data used	Epochs	Update seed	Prediction error Δ_2		Planning error Δ_3		Statistically significant increase
					old-pipeline training	isolated training	old-pipeline training	isolated training	
1	0	100%	10	1	+0.049	−0.141	−0.027	+0.035	none
2	0	100%	10	2	+0.062	−0.212	+0.026	+0.067	none
3	0	100%	10	3	+0.139	−0.060	+0.020	+0.078	none
4	0	50%	10	1	+0.184	+0.021	−0.012	+0.081	none
5	0	50%	10	2	+0.192	+0.100	+0.031	+0.063	none
6	0	50%	10	3	+0.098	−0.154	+0.040	+0.127	none
7	0	25%	10	1	+0.233	−0.058	−0.030	+0.020	none
8	0	25%	10	2	+0.284	+0.082	−0.042	+0.071	prediction
9	0	25%	10	3	+0.157	−0.141	−0.046	+0.034	none
10	0	100%	5	1	+0.182	−0.073	−0.024	+0.052	none
11	0	100%	5	2	+0.284	+0.001	−0.006	+0.040	prediction
12	0	100%	5	3	+0.266	+0.093	−0.042	+0.033	prediction
13	0	100%	20	1	+0.107	−0.079	−0.000	+0.051	none
14	0	100%	20	2	+0.127	−0.051	+0.025	+0.059	none
15	0	100%	20	3	+0.115	−0.085	+0.001	+0.056	none
16	1	100%	10	1	−0.054	−0.170	−0.001	+0.021	none
17	1	100%	10	2	−0.081	−0.156	+0.026	+0.073	none
18	1	100%	10	3	−0.187	−0.280	+0.023	+0.026	none
19	1	50%	10	1	−0.030	−0.115	+0.072	+0.084	planning
20	1	50%	10	2	−0.135	−0.152	+0.054	+0.088	none
21	1	50%	10	3	−0.089	−0.141	+0.037	+0.070	none
22	1	25%	10	1	+0.090	−0.059	−0.001	+0.029	none
23	1	25%	10	2	+0.081	−0.095	−0.042	−0.004	none
24	1	25%	10	3	+0.082	−0.111	+0.010	+0.022	none
25	1	100%	5	1	−0.025	−0.267	+0.033	+0.068	none
26	1	100%	5	2	+0.040	−0.094	+0.003	+0.064	none
27	1	100%	5	3	+0.052	−0.107	+0.015	+0.050	none
28	1	100%	20	1	+0.003	−0.094	+0.049	+0.047	none
29	1	100%	20	2	+0.065	−0.029	+0.021	+0.022	none
30	1	100%	20	3	+0.073	−0.025	+0.018	+0.049	none

Bottom line

- Entangled enhancement is real for the prediction module: a better C_1 harms the C_2 that learned the old C_1, and not the C_2 trained on the ground truth.
- The effect on planning is very small.
- Retraining C_2 and C_3 together with C_1 lowers both errors, but no cheap warning sign has been found yet. Early detection is the next step.